RelaLeap: What Was Actually Tested in the Predictive-Coding / ePC Track?

Consolidated experiment report
RelaLeap project record

6 August 2026

Executive summary

This report corrects an important naming issue in the earlier experiment history. The runs originally called "ePC" (especially R6–R9) used a *state-based* predictive-coding implementation: hidden states were optimized and then detached before parameter differentiation. A later source and method audit established that this is SO/sPC, not the error-variable EO/ePC method described in the reference paper. Therefore their negative performance results are evidence against that *legacy implementation*, not a clean test or refutation of genuine EO/ePC.

The legacy programme did produce one robust positive engineering observation: its declared activity-energy decreased monotonically in 10/10 multi-seed records. But it did not turn that property into practical advantages. On a six-layer GPT-2 student trained on WikiText-103, ordinary backpropagation with knowledge distillation (KD), given the same elapsed time, made about 6.7 times as many updates and had substantially better validation loss. Legacy PC also showed lower representation rank and weaker causal-factor selectivity.

The subsequent four-regime CPU preflight likewise does *not* establish an ePC efficacy result: its ordinary T=1 control could not learn the fixture. Moreover, the later audit found a metric defect that selected the worst rather than the best cap. Its old arm comparisons are invalid for method comparison.

The replacement EO/ePC implementation has passed local mechanism and instrument checks: it has direct error-variable dynamics, settles to the exact PC fixed point in small analytical fixtures, has a validated target/evaluation seam, and replays bit-exactly. This is validity evidence, not yet a language-model efficacy result. The planned five-seed GPT-2/WikiText reference reproduction (C1) has not successfully run; it remains blocked pending a fresh, single-writer GPU execution envelope.

1 How to read the evidence

- *Observed* means an artifact or recorded result directly supports the statement.
- *Inference* distinguishes what the results suggest from what they prove.
- A completed software or analytic control is not treated as evidence that EO/ePC improves language-model learning.

2 Chronology and status

Date	Run	Question	Result / status
19 Jul	R6 mechanism screen	Can five arms complete and expose dynamics?	Failed after the two BP arms; no PC result.
20 Jul	R7 one-seed screen	Does the repaired five-arm legacy run complete?	Completed; small loss edge over update-matched KD, no benefit from deeper inference. Exploratory only.
20–21 Jul	R8, five seeds	Does legacy PC beat BP/KD on GPT-2/WikiText?	No: BP-KD at matched time dominated; energy monotonicity held for legacy PC.

Date	Run	Question	Result / status
21–22 Jul	R9 representation battery	Does legacy PC yield better internal representations?	No: lower rank and lower causal selectivity than BP-CE.
4 Aug	Multiregime preflight	Is a larger planted-task fixture learnable before a frontier/MoE study?	No: the matched T=1 control failed every regime; no frontier study was admitted. Later audit also invalidated its old comparison metric.
5 Aug onward	EO/ePC remediation	Is the method and measurement apparatus actually EO/ePC and trustworthy?	Local method, fixed-point, target-seam, and evaluator controls passed. No language efficacy result yet.

3 Legacy experiments (R6–R9)

3.1 R6: interrupted mechanism screen

R6 was a one-seed, 200-update GPU screen on a six-layer GPT-2 student and WikiText-103. It compared BP cross-entropy (CE), update-matched BP-KD, two legacy PC depths (T=4 and T=12), and wall-clock-matched BP-KD. The job failed after CE and KD completed, before any legacy-PC arm completed. Its immediate exception was not retained. Thus it supports no PC conclusion. The two saved baselines finished at validation losses 8.439 (CE) and 7.628 (KD).

3.2 R7: repaired exploratory screen

R7 fixed the arm-loop crash and completed all five arms for one seed. At 200 updates the reported validation losses were: CE 8.4390, KD 7.6278, legacy PC T=4 7.5672, legacy PC T=12 7.5836, and wall-clock KD 7.6278. This was an interesting single-seed hint, but it was not a confidence interval or a decisive comparison. Increasing inference depth fourfold did not help.

3.3 R8: five-seed outcome comparison

R8 repeated the design across five frozen seeds (25 records). It is the main legacy performance result.

Arm	mean validation loss	mean perplexity	updates	elapsed seconds	interpretation
BP-CE	6.261 ± 0.060	524	200	78	best equal-update baseline
BP-KD	7.039 ± 0.056	1142	200	74	KD noisy at 200 updates
Legacy PC T=4	6.781 ± 0.146	889	200	445	slower, worse than CE
Legacy PC T=12	6.754 ± 0.131	863	200	1738	no convincing depth gain
BP-KD, matched time	5.785 ± 0.135	328	1334	445	dominant matched-time arm

Observed. All 10 legacy-PC seed/arm records satisfied the declared activity-energy-monotonicity check. In the time taken by T=4 legacy PC to make 200 updates, ordinary KD made 1334: approximately 6.7 times more.

Inference. The evidence says that the legacy state-PC implementation did not provide a useful speed/quality trade-off on this task. It does not establish that EO/ePC would behave identically, because the algorithm was not the same.

3.4 R9: representation and causal-factor diagnostics

R9 analyzed 20 of the R8 trained models. At the final layer, effective rank was 13.9 ± 0.3 for BP-CE, 5.0 ± 0.1 for BP-KD, 3.9 ± 0.2 for legacy PC T=4, and 3.8 ± 0.1 for legacy PC T=12. Participation ratio was 6.1 for CE, 3.1 for KD, and about 2.6 for both legacy-PC arms. These are strong signs of representational compression, not dead units (dead fraction was 0.001).

Final-layer causal-selectivity probes also favored BP-CE. For subject number, object number, tense, and negation respectively, BP-CE scored 0.607, 0.425, 0.661, and 0.870; legacy T=4 scored 0.438, 0.232, 0.546, and 0.719;

legacy T=12 scored 0.379, 0.275, 0.481, and 0.718. The original interpretation was that smoother activity energy coincided with more compressed but less factor-selective representations. That remains a reasonable description of these legacy runs, but not a general theorem about EO/ePC.

4 The multiregime frontier preflight

The four-regime CPU preflight was meant to admit a richer planted transformer task before testing a Pareto frontier of PC variants or an MoE-like conditional specialization scheme. Twelve arm-by-seed executions replayed exactly; T=2 and T=4 were energy-monotone. But the required T=1 control had NLLs far above its 1.4 learnability floor in every regime (roughly 6.76–11.92). The fixture was therefore correctly stopped before a frontier claim.

Caveat. A later source audit established both that legacy code was SO/sPC and that a negation in the old cap-selection path selected the worst, rather than best, cap. Hence the values from this preflight cannot be used to compare methods. What survives is the narrow operational fact: the fixture, as then specified, failed the ordinary-control learnability gate.

5 EO/ePC remediation: what has now been established

After the method audit, the track was rebuilt around the paper's error-variable computation. The following are local, reproducible validation steps.

Gate	Observed result	Meaning and limit
A0 method/instrument audit	Pinned paper and official-code sources matched; uniform NLL was ln(64); planted-oracle NLL was about 2.38×10^{-9}; trainer-teacher round trip was 0.0.	Established old routine was SO/sPC and validated basic measurement paths.
MG-1 parameterization	At depths 2/6/12/24, every first error update equaled the output adjoint times -0.01 (relative error 0). Old state-based path had zero first-step update in earlier layers.	Establishes direct deep error signal in the new EO/ePC core; no efficacy claim.
MG-2R fixed-point check	All 12 EO points settled (relative residual 5.53×10^{-9} to 8.14×10^{-9}), matched exact controls, and replayed bit-exactly.	EO/ePC realizes the finite-lambda PC fixed point in the tested analytical fixtures.
MG-3/MG-4 settling/stability	Explicit residual certificates and local map-stability estimates passed their frozen controls.	Shows numerical diagnostics behave as declared, only for the tested maps.
MG-5/MG-6 trainer seam	All 512 planted target positions had a unique correct target; corruptions failed closed; evaluation/teacher round trips passed.	Target production and scoring are no longer the known source of a false conclusion.

One earlier MG-2 formulation failed because it demanded finite-lambda equality to BP at tolerance 10^{-6}, which a seed-free scalar counterexample showed to be mathematically impossible. MG-2R replaced that invalid demand with the correct lambda-to-zero convergence-rate criterion and passed. This is a repair of a test contract, not post-hoc tuning of an efficacy outcome.

6 What is known now, and what is not

Known. The old state-PC experiments were completed and their raw outcomes are preserved. They show monotone declared activity energy but no competitive learning or representation benefit against their baselines. The new core is materially different: it has passed analytic EO/ePC mechanism and measurement controls.

Not known. Whether genuine EO/ePC improves a transformer on language-model learning, distillation, robustness, adaptation, or causal representation quality is still untested. The relevant next study is the frozen C1 five-seed reproduction with GPT-2 and WikiText. Its local package and adjudication seams have been built, but no valid C1 science run has completed; prior remote attempts were terminated before model/data/seed work because of package and provider-control failures.

Primary evidence and reproducibility

The project ledgers are authoritative. This report is a synthesis, not a replacement for raw artifacts.

- `experiments/20260719T202100Z-epc-mechanism-screen-r6/RUN.md`
- `experiments/20260720T114200Z-epc-mechanism-screen-r7/RUN.md`
- `experiments/20260720T193211Z-epc-r8-broad-outcome/RUN.md`
- `experiments/20260721T222400Z-r9-stage-a/RUN.md`
- `experiments/20260804T173241Z-epc-multiregime-frontier-preflight-v1-r5/RUN.md`
- `experiments/20260805T002848Z-epc-remediation-a0-acceptance-r2/RUN.md`
- `experiments/20260805T004556Z-epc-remediation-mg1-artifact-acceptance/RUN.md`
- `experiments/20260805T015749Z-epc-remediation-mg2r-acceptance/RUN.md`
- `experiments/20260805T035047Z-epc-remediation-mg5-acceptance/RUN.md`

The earlier historical companion is `docs/prior_epc_experiment_history_20260804.pdf`; this report supersedes its method-identity interpretation where the later audit found the SO/sPC versus EO/ePC distinction.